

administrative platform engineered specifically for East African healthcare facilities. It digitises every touchpoint of a patient's journey — from triage and outpatient consultation through inpatient ward rounds, pharmacy dispensing, laboratory results, imaging review, billing, and discharge — while remaining compliant with Kenya's Data Protection Act 2019 and connected to national reporting infrastructure (DHIS2/KHIS and the SHA/SHIF insurance scheme).

Built on a modern, containerised Python/Flask architecture backed by PostgreSQL, the system has been hardened through six development phases, stress-tested at sustained concurrency, and prepared for independent clinical safety and security audits. With 214 automated tests, a 41.6% code-coverage baseline, and a 0.0% error rate on all hot-path benchmarks, the platform is engineered to enterprise reliability standards.

214 Automated tests passing	41.6% Code coverage baseline	0.0% Hot-path error rate	17 Clinical & admin modules	6 Phases completed
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2. Clinical Modules

Module	Key Capabilities
Outpatient / OPD	Patient registration with UUID, triage vitals capture, OPD waiting-list queue, SOAP/SBAR consultation notes, ICD-10 diagnosis coding (50+ curated codes), AI Clinical Assistant (chatbot with audit trail & consent gate).
Ward Rounds & Inpatients	Admit/discharge workflow, ward-bed history, daily ward round notes, Medication Administration Record (MAR), nurse notification system.
Prescribing & CDSS	Drug prescribing with Clinical Decision Support: 7,621 live drug–drug interaction rules via DrugCentral PostgreSQL, allergen class screening (5 classes), renal dose adjustment, Patient Global Allergy Registry, Active Problem List.
Pharmacy	Dispensing workflow, FEFO (First-Expiry-First-Out) batch inventory, automated low-stock scanning, auto-generated Supplier Purchase Orders, AI drug-discovery assistant, cheminformatics fingerprint search.
Laboratory	Lab test ordering from consultation, LIS result entry, panic-value alert system (LIS blueprint), result gating before patient-portal release.
Imaging / Radiology	DICOM study upload (up to 2 GB), Cornerstone.js in-browser viewer, unmatched-imaging reconciliation queue, PACS/HL7 interfacing research documented.
Theatre & Oncology	Theatre booking and list management, post-operative note update, Oncology clinic with chemotherapy regimen tracking, AI treatment summary (consent-gated).
Emergency Access	Break-glass emergency override with full audit trail, time-limited access tokens, supervisor alert dispatching, admin audit-trail view.
Telemedicine	WebRTC virtual consultation room with real-time signalling via Socket.IO, in-call clinical notes, prescription drafting. Feature-flagged (ENABLE_TELEMEDICINE) pending regulatory sign-off.

3. Administrative & Operational Modules

Module	Key Capabilities
Billing & Finance	Invoice generation, multi-payment allocation (cash, M-Pesa STK Push, insurance), unreconciled-charges aggregation, revenue analytics.
M-Pesa Integration	Safaricom Daraja API STK Push and C2B callback handling, payment receipting, patient-portal self-pay flow.
SHA/SHIF Insurance	InsuranceScheme & PatientInsurance models, claim adjudication workflow, approval/rejection tracking, analytics breakdown.
HR & Credentialing	Staff roster management, professional-licence upload, expiry-date warning alerts, StaffCredential model with admin view.
Stores & Inventory	Central stores issue/return workflow, stock-level tracking, reorder triggers.
Mortuary	Deceased-patient registration, body release workflow.
Analytics Dashboard	Executive dashboard (Chart.js): bed occupancy trend, 30-day admission curve, revenue breakdown by payment channel, insurance claim approval ratios. JSON API for BI tool integration.
Patient Self-Service Portal	Separate PatientUser authentication, appointment booking, lab-result viewing (gated), billing history, M-Pesa self-pay, profile management with audit log.
Outbound Communications	Flask-Mail email driver, SMS sandbox abstraction, OutboundNotificationLog, 5 event triggers (appointments, labs, billing, payments, claims), 24-hour appointment reminder scheduler.
Audit & Logging	AuditLog DB model, @audited decorator on all write routes, SIEM-export endpoint, full structured event log with user/IP/timestamp/diff.

4. Interoperability & Standards

Module	Key Capabilities
FHIR R4 API	RESTful /api/fhir/R4 endpoints for Patient, Observation, MedicationRequest, DiagnosticReport resources. Enables integration with national health exchanges.
DHIS2 / KHIS Export	/api/khis blueprint: automated aggregate report generation and push to Kenya's national DHIS2 instance for MOH reporting compliance.
JWT REST API	Bearer-token authenticated /api/* endpoints for mobile clients, third-party EHR connectors, and BI dashboards.
M-Pesa (Daraja)	STK Push initiation and C2B webhook for real-time payment reconciliation.
ICD-10 Coding	50+ curated ICD-10-CM codes across all departments; WHO API credential requirement documented for full 70,000-code database ingestion.
DICOM / PACS	DICOM Web upload & Cornerstone.js viewer. HL7 MLLP/ASTM LIS interfacing architecture documented; implementation pending vendor selection.

5. Security & Compliance

Security is implemented as a layered defence-in-depth strategy across system layers:

Module	Key Capabilities
Authentication	TOTP-based MFA, 5-attempt account lockout (15-minute timeout), bcrypt/PBKDF2 password hashing, session signed with SECRET_KEY.
Authorisation	Role-based access control (RBAC) across 12 roles; @break_glass_required decorator for emergency overrides with full audit trail.
Encryption at Rest	Fernet AES-128-CBC + HMAC EncryptedString column type applied to all patient PII identity fields.
Data Protection Act 2019	PatientConsent model, Subject Access Request JSON export, patient anonymisation route, data-residency hard stop documented.
CSRF Protection	Flask-WTF CSRF tokens on all state-changing forms; JWT API endpoints explicitly exempted.
Rate Limiting	Flask-Limiter on login (5/min POST), disabled in test mode.
CI Security Gates	pip-audit (17 known advisories documented), bandit SAST, import-order linting — all enforced in GitHub Actions without bypasses.
Session Security	Redis-backed sessions, 30-minute idle timeout, SameSite=Lax, HttpOnly, Secure cookie flags.

6. Technical Architecture

6.1 Stack Overview

Module	Key Capabilities
Backend	Python 3.12 · Flask 3.x · SQLAlchemy ORM · Flask-Migrate (Alembic)
Database	PostgreSQL 16 (primary) · SQLite (test / offline fallback)
Cache / Queue	Redis 7 (session store + Celery broker) · Celery 5.4 (async tasks)
Real-time	Flask-SocketIO · WebRTC (telemedicine signalling)
Auth	Flask-Login · Flask-JWT-Extended · PyOTP (TOTP MFA)
AI / NLP	NVIDIA NIM API client · local HuggingFace summariser · DrugCentral DDI DB
Containers	Docker + Docker Compose (web, db, redis, celery_worker, nlp services)
CI/CD	GitHub Actions: lint → bandit → pip-audit → pytest (214 tests, 41.6% cov)
Offline / PWA	Service Worker sw.js · Web App Manifest · offline.html fallback

6.2 Performance Baseline

Hot-path benchmarking was executed with a session-authenticated concurrent load test (C = 20 simultaneous workers). Results recorded in docs/load_test_results.md:

Module	Key Capabilities
Patient Registration	0.0% error rate · sub-200 ms p95 at C=20
Prescription Sign-off	0.0% error rate · sub-250 ms p95 at C=20
Billing Payment	0.0% error rate · sub-200 ms p95 at C=20
Health Check /healthz	1,200 req/sec sustained · Flask-Limiter lockout verified

7. Audit & Regulatory Readiness

Three independent audit-preparation packages are maintained within the repository:

[docs/security_audit_readiness.md](#)

Maps OWASP Top-10 controls to implementation routes. Identifies 17 open pip-audit advisories with risk ratings and mitigations.

[docs/clinical_safety_review_packaging.md](#)

Documents Clinical Decision Support rules, AI governance framework (input validation, consent gating, audit timer), and allergy registry.

[docs/accessibility_audit_report.md](#)

WCAG 2.1 AA compliance verification covering keyboard navigation, ARIA landmarks, and screen-reader compatibility.

8. Recommended Next Steps

Module	Key Capabilities
ICD-10 Full Database	Expand 50-item curated array to full ICD-10-CM FTS5 index via WHO API.
KRA eTIMS Middleware	Implement tax middleware for automated KRA electronic invoicing.
Controlled Drugs	Schedule I/II/III dispensing log with dual-signature authorization.
Hardware Interfacing	ASTM E1381 MLLP listener for lab analyzers and PACS DICOM worklists.
Penetration Testing	Third-party OWASP web application penetration audit prior to go-live.

9. Contact & Licensing

This document is confidential and intended solely for designated stakeholders. The HMIS codebase is maintained under a proprietary license. For deployment inquiries, clinical configuration, or integration support, contact the engineering team via the repository portal.

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